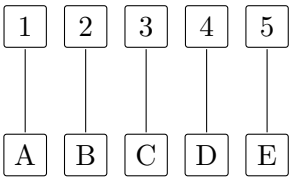
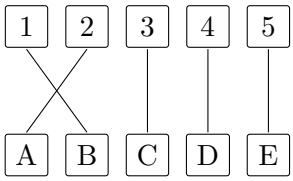
There are two classes of kids, and each class lines up. A complete match gives each kid a partner in the other class, with nobody left out. It keeps order when any two kids in the first class appear in the same order as their partners in the second class.

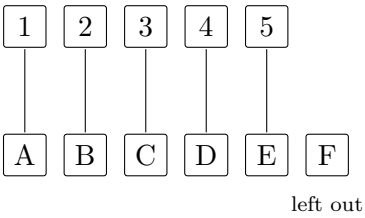
Complete match keeps order



Complete match does not keep order



5 kids and 6 no complete match possible



Problem 1: Find every complete match between these two class lines. Which matches keep order?

R1	R2	R3
B1	B2	B3

R1	R2	R3	R4
R5	R6	R7	R8
R9	R10	R11	R12
B1	B2	B3	B4
B5	B6	B7	B8
B9	B10	B11	B12